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Micro-Doppler feature plays an important role in cone-shaped target recognition and parameter estimation. However, translational motion causes shifting, tilting, and aliasing in micro-Doppler spectrum and influences the extraction of micro-Doppler features. Most existing methods do not consider higher order translation compensation. In this article, a high-order translational motion compensation method is proposed. First, the high-order ambiguity function (HAF) of a periodical polynomial phase signal exhibits periodicity along the lag axis and reaches the peak concentration when the lag is equal to the micro-motion period. The micromotion period can be obtained by searching the peak concentration of echo's HAF. Then, the translational jerk and acceleration are, respectively, compensated by searching the peaks of the second- and third-order HAFs. Last, convolution is operated on the frequency spectrum to find the symmetric center located at the Doppler shift frequency, and the residual translational velocity can be estimated. Experimental results based on electromagnetic computation data show that the proposed method can accurately achieve high-order translational compensation.

1. INTRODUCTION

Micromotion dynamics can induce Doppler modulations on the returned echoes, which are called micro-Doppler effect [1], [2]. The micromotion feature can be extracted by the time-Doppler analysis of the radar echoes [3], [4], [5], [6], [7], [8]. The space targets, such as the space debris and the space vehicles, usually have the high-speed maneuverable translational motion. In real scenarios, the translational motion between the target and the radar, which also induces the Doppler modulation on echoes, can cause the shifting, tilting, and aliasing in micro-Doppler spectrum. Thus, translational motion influences the extraction for micromotion features, and its compensation is a necessary preprocess.

Many articles have been conducted on translational motion compensation. In a previous article [9], a method based on the minimization of Doppler spectrum entropy is proposed to estimate translational parameters. However, the

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search interval and scope, which influence estimation accuracy and computational complexity, are hard to determine. The article [10] proposes a method based on empirical mode decomposition, which separates a translational component based on differences between translational Doppler and micro-Doppler frequencies. However, frequency aliasing in the time-frequency (TF) domain makes this method invalid. Furthermore, Radon transform is used to detect the inclination of TF curves [11]. Then, translational parameters are estimated by searching peaks in the TF spectrum with a minimum entropy criterion and Gaussian function fitting. However, this method has poor noise robustness, and it is limited by the resolution of the TF spectrum. A novel compensating method for a rotationally symmetric target is proposed by utilizing the local symmetry cancellation (LSC) effect [12]. Gu et al. [13] estimated the translational acceleration by delayed-conjugated multiplication (DM) processing, and residual velocity is compensated by searching the maximum spectral peak. Nevertheless, this method is only suitable for the sinusoidal modulated echoes of spinning targets. In [14] and [15], the ambiguity function (AF) is used to estimate the motion period. AF is free from cross-term interference because it does not rely on the TF representation. However, these methods are based on the assumption of quadratic translation and do not consider cubic or higher order translation.

In this article, we propose a translational motion compensation method for a precession cone-shaped target with cubic translation. We can regard the echoes of the target as a periodical polynomial phase signal because of precession motion with cubic translation. We transform the translational motion compensation problem as a polynomial phase signal parameter estimation problem. First, with the analytical formula, we prove that the high-order ambiguity function (HAF) [16] of the periodical polynomial phase signal exhibits periodicity along the lag axis, and reaches the peak concentration when the lag is equal to the micromotion period. We use the HAF entropy to describe the energy concentration, and obtain the micromotion period by determining the peak of the entropy. Then, we compensate the translational jerk and acceleration by searching the peaks of second- and third-order HAFs, respectively. Last, since the target undergoes periodical motion, the support area of the frequency spectrum is symmetric around the Doppler shift frequency caused by translational velocity. We operate the convolution on the frequency spectrum to determine the symmetric center and estimate the residual translational velocity. The main contributions of this article are as follows:

- 1) We propose a method for high-order translation motion compensation. With the detailed analytical formula, the HAF of the periodical polynomial phase signal is proven periodical along the lag axis and reaches the peak concentration when the lag is equal to the micromotion period. Electromagnetic experimental results demonstrate that the proposed method

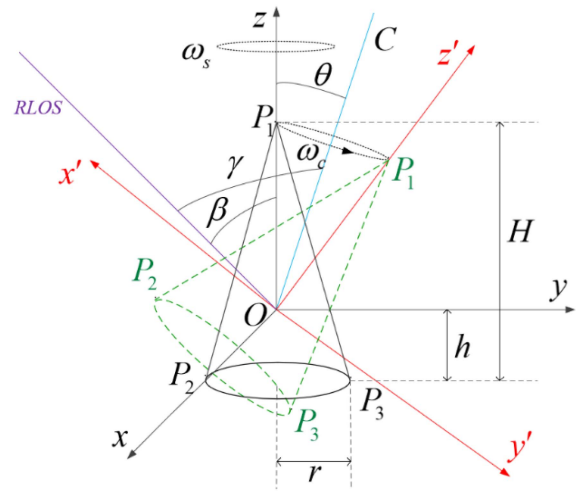


Fig. 1. Micromotion model of the cone-shaped target.

can accurately estimate parameters in the case of high-order translation.

- 2) We propose a translational velocity compensation method by the symmetry of the frequency spectrum. This method operates the convolution without determining the search interval and search scope.

The organization of this article is as follows. The geometric model and precession model of a cone-shaped target are given in Section II. The translational motion compensation based on HAF and spectral information is presented in Section III. The performance evaluation with experimental results is shown in Section IV. Section V concludes this article.

II. PRECESSION MODEL OF A CONE-SHAPED TARGET

The micromotion model of a cone-shaped target is shown in Fig. 1. $(O - xy'z')$ is the reference coordinate system, and $(O - xyz)$ is the target coordinate system. θ , ω_s , and ω_c are the precession angle, spin angular velocity, and precession angular velocity, respectively, used to describe precession motion. The angle between the radar line-of-sight (RLOS) and the precession axis OC is γ , and the target line-of-sight angle between RLOS and Oz is β . According to a previous article [17], the microrange of scattering centers can be described as

$$\begin{cases} r_{m1}(t) = -(H - h) \cos \beta(t) \\ r_{m2}(t) = h \cos \beta(t) - r \sin \beta(t) \\ r_{m3}(t) = h \cos \beta(t) + r \sin \beta(t) \end{cases} \quad (1)$$

where h is the microrange from the bottom plane to the origin O ; and H and r are the height and radius of the target, respectively. Variation in β can be expressed as follows:

$$\cos \beta(t) = \cos \gamma \cos \theta - \sin \gamma \sin \theta \cos(\omega_c t + \varphi_0) \quad (2)$$

where φ_0 is the initial phase. In (2), micromotion by precession is periodic with a period of $T_m = 2\pi/\omega_c$.

The translational range in a short dwell time is usually approximated by a cubic polynomial, which can be obtained

by

$$r_T(t) = r_0 + v_0 t + a_1 t^2 / 2 + a_2 t^3 / 6 \quad (3)$$

where r_0 is the initial range; and v_0 , a_1 , and a_2 are the translational velocity, acceleration, and jerk, respectively.

If the narrowband radar transmits a single frequency signal [18], the baseband signal can be expressed as

$$s(t) = \sum_{i=1}^3 \sigma_i \exp[-j4\pi (r_{im}(t) + r_T(t)) / \lambda] = s_m(t) s_T(t) \quad (4)$$

where

$$\begin{aligned} s_m(t) &= \sum_{i=1}^3 \sigma_i \exp(-j4\pi r_{im}(t) / \lambda) \\ s_T(t) &= \exp(-j4\pi r_T(t) / \lambda) \end{aligned} \quad (5)$$

where σ_i is the scattering coefficient of the scattering center P_i ; λ is the echo wavelength; and $s_m(t)$ and $s_T(t)$ are the signal components for micromotion and translational motion, respectively.

III. TRANSLATIONAL MOTION COMPENSATION BASED ON HAF AND SPECTRAL INFORMATION

According to (4), the echo of the precession target with translational motion can be regarded as a periodical polynomial phase signal. In this article, the translational motion compensation problem is transformed as a polynomial phase signal parameter estimation problem. First, the micromotion period is estimated on the basis of the HAF energy concentration. Then, jerk and acceleration are estimated with second- and third-order HAF. Last, a translational velocity compensation method is proposed on the basis of frequency spectral symmetry.

A. Micromotion Period Estimation

The M th-order AF [16] can be defined as the Fourier transform of the M th-order instantaneous moment with respect to the variable t

$$R_M(f, \tau) = \int_{-\infty}^{+\infty} c_M(t, \tau) \exp(-j2\pi f t) dt. \quad (6)$$

The M th-order instantaneous moment is described as

$$c_M(t, \tau) = \prod_{q=0}^{M-1} [s^{\dagger q}(x + q\tau)]^{C_{M-1}^q} \quad (7)$$

where

$$s^{\dagger q}(x) = \begin{cases} s(x) & q \text{ is even} \\ s^*(x) & q \text{ is odd} \end{cases} \quad (8)$$

where $(\cdot)^*$ is the complex conjugation, τ is the lag, and C_y^x is in the form of $C_y^x = y! / (y-x)! x!$.

According to (3) and (4), the echoes have third-order phase. Therefore, we choose third-order AF. When $M = 3$,

the third-order instantaneous moment $c_3(t, \tau)$ can be expressed as

$$\begin{aligned} c_3(t, \tau) &= s(t) [s^*(t + \tau)]^2 s(t + 2\tau) \\ &= [s_m(t) [s_m^*(t + \tau)]^2 s_m(t + 2\tau)] \cdot \\ &\quad [s_T(t) [s_T^*(t + \tau)]^2 s_T(t + 2\tau)] \\ &= c_{3m}(t, \tau) c_{3T}(t, \tau) \end{aligned} \quad (9)$$

where $c_{3m}(t, \tau)$ and $c_{3T}(t, \tau)$ are the third-order instantaneous moments of translational and micromotion components, respectively.

In this article, the HAF of the periodical polynomial phase signal exhibits periodicity along the lag axis and reaches the peak concentration when the lag is equal to the micromotion period. First, we discuss the periodicity of the micromotion component's HAF. Let $\tau = \tau + T_m$. Due to the micromotion signal component $s_m(t)$ is periodical, we can obtain the following formula:

$$\begin{aligned} c c_{3m}(t, \tau + T_m) &= s_m(t) [s_m^*(t + \tau + T_m)]^2 s_m(t + 2\tau + 2T_m) \\ &= s_m(t) [s_m^*(t + \tau)]^2 s_m(t + 2\tau) \\ &= c_{3m}(t, \tau). \end{aligned} \quad (10)$$

According to (6), the third-order AF of micromotion component $s_m(t)$ satisfies

$$R_{3m}(f, \tau + T_m) = R_{3m}(f, \tau). \quad (11)$$

After that, we discuss the HAF of the translational component, which is expressed as

$$\begin{aligned} R_{3T}(f, \tau) &= \int_{-\infty}^{+\infty} c_{3T}(t, \tau) \exp(-j2\pi f t) dt \\ &= \int_{-\infty}^{+\infty} \exp(-j4\pi(a_1 \tau^2 + a_2 \tau^2 t + a_2 \tau^3) / \lambda) \\ &\quad \times \exp(-j2\pi f t) dt \\ &= \delta(f + a_2 \tau^2 / \lambda) \exp(-j4\pi(a_1 \tau^2 + a_2 \tau^3) / \lambda). \end{aligned} \quad (12)$$

Combining (6), (11), and (12), we can obtain the HAF of target echoes as

$$\begin{aligned} R_3(f, \tau) &= R_{3m}(f, \tau) \otimes R_{3T}(f, \tau) \\ &= \exp(-j4\pi(a_1 \tau^2 + a_2 \tau^3) / \lambda) R_{3m}(f + a_2 \tau^2 / \lambda, \tau) \end{aligned} \quad (13)$$

where $\otimes$ denotes the convolution operator. From (13), we can find that the energy of HAF is only depended on the micromotion component. The third-order instantaneous moment of micromotion component can be rewritten as

$$\begin{aligned} c_{3m}(t, \tau) &= s_m(t) [s_m^*(t + \tau)]^2 s_m(t + 2\tau) \\ &= \sum_{i=1}^3 \sigma_i^4 \exp\left(-j \frac{4\pi}{\lambda} (r_{im}(t) - 2r_{im}(t + \tau) \right. \\ &\quad \left. + r_{im}(t + 2\tau))\right) + C_{3r}(t, \tau). \end{aligned} \quad (14)$$

We can find that the instantaneous moment calculates the cross-correlation of $s(t)$, $s(t + \tau)$, and $s(t + 2\tau)$. Since the micromotion component is periodical, when $\tau = T_m$, the phase difference of these three signals is the smallest, and the energy of HAF is most concentrated.

Then, we analyze the HAF energy concentration with the analytical formula. The autoterm of $c_{3m}(t, \tau)$ is equal to 1 when $\tau = T_m$, and we have

$$R_{3m}(f, T_m) = \delta(f) \quad (15)$$

where $\delta(f)$ is the Dirac delta function. Additionally, the cross-terms in the frequency domain are dispersive. From (15), we can find that the energy of HAF concentrates at zero frequency when $\tau = T_m$. According to the analysis discussed above, we can estimate the period by finding the peak location of HAF energy concentration. Usually, the energy concentration of a signal is described by the signal's entropy, such as TF entropy [19] and ISAR imaging entropy [20]. Therefore, we use the HAF entropy to describe the HAF energy concentration.

In the following, we give the analysis for HAF energy concentration with the analytical formula. The HAF entropy, which is the entropy of $R_3(f, \tau)$ along the frequency dimension, can be defined as

$$\varepsilon(\tau) = \int_{-\infty}^{\infty} \frac{|R_3(f, \tau)|}{E(\tau)} \ln \frac{|R_3(f, \tau)|}{E(\tau)} df \quad (16)$$

where

$$E(\tau) = \int_{-\infty}^{\infty} |R_3(f, \tau)| df. \quad (17)$$

Substituting (13) into (16) and (17), we can obtain

$$\begin{aligned} \varepsilon(\tau) &= \int_{-\infty}^{\infty} \frac{|R_3(f, \tau)|}{E(\tau)} \ln \frac{|R_3(f, \tau)|}{E(\tau)} df \\ &= \int_{-\infty}^{\infty} \frac{|R_{3m}(f', \tau)|}{E(\tau)} \ln \frac{|R_{3m}(f', \tau)|}{E(\tau)} df' = \varepsilon_m(\tau) \end{aligned} \quad (18)$$

where $f' = -a_2\tau^2/\lambda$, and $\varepsilon_m(\tau)$ is the HAF energy concentration of the micromotion component. Since $R_{3m}(f, \tau)$ is periodical and peaks at $\tau = T_m$, the periodicity and peak location of the HAF energy concentration is proven. In summary, the HAF of the periodical polynomial phase signal exhibits periodicity along the lag axis and reaches the peak concentration when $\tau = T_m$. The micromotion period can be obtained by finding the peak of the HAF energy concentration.

B. Jerk and Acceleration Compensation

According to (15), we can find that the third-order HAF of the translational component peaks when $\tau = T_m$. Therefore, we can estimate the jerk by

$$\hat{a}_2 = -\lambda f_{\text{peak}} / 2\tau^2 \quad (19)$$

where f_{peak} is the peak of $S_3(f, T_m)$.

After the jerk phase in (4) is compensated, the echo can be expressed as

$$s'(t) = \sum_{i=1}^3 \sigma_i \exp \left[-j4\pi (r_{im}(t) + r_0 + v_0 t + a_1 t^2 / 2) / \lambda \right]. \quad (20)$$

Similarly, we use the second-order HAF to estimate acceleration. The second-order instantaneous moment of $s'(t)$ can be written as

$$\begin{aligned} s'_2(t, \tau) &= s'(t) s'^*(t + \tau) \\ &= T_2(t, \tau) (A_{2m}(t, \tau) + C_{2m}(t, \tau)) \end{aligned} \quad (21)$$

where

$$\begin{aligned} T_2(t, \tau) &= \exp(j4\pi(v_0\tau + a_1t\tau + a_1\tau^2/2)/\lambda) \\ A_{2m}(t, \tau) &= \sum_{i=1}^3 \sigma_i^2 \exp(-j4\pi(r_{im}(t) - r_{im}(t + \tau))/\lambda) \end{aligned} \quad (22)$$

where $A_2(t, \tau)$ and $C_2(t, \tau)$ are the autoterms and cross-terms of a micromotion component. Similar to the analysis of the third-order HAF, the peak of $S'_2(f, T_m)$ is located at $f'_{\text{peak}} = 2a_1\tau/\lambda$, and acceleration can be estimated using

$$\hat{a}_1 = \lambda f'_{\text{peak}} / 2\tau. \quad (23)$$

C. Velocity Compensation

After compensating the translational jerk and acceleration, we obtain the radar echo form as

$$\tilde{s}(t) = \sum_{i=1}^3 \sigma_i \exp \left[-j \frac{4\pi}{\lambda} (r_{im}(t) + r_0 + v_0 t) \right]. \quad (24)$$

In a period dwell time, the frequency spectrum of the radar echo is approximately symmetrical at zero frequency if translation motion is completely compensated. Therefore, the support area of the frequency spectrum of (29) is symmetric at the Doppler shift frequency. We use the convolution operator to find the symmetry center and estimate the velocity. The frequency spectrum of $\tilde{s}(t)$ can be expressed using the following equation by performing Fourier transform to the signal $\tilde{s}(t)$:

$$F(k) = \sum_{n=0}^{N-1} \tilde{s}(n) \exp(-j2\pi kn/N) \quad (25)$$

where n and k are the indices of slow time and Doppler frequency, respectively, and N is the length of slow time. Then, we perform the convolution operator by using

$$\begin{aligned} P(M) &= |F(k) \otimes F(k)| = \left| \sum_{k=-N/2}^{N/2} F(k) F(M-k) \right| \\ \hat{M} &= \underset{M}{\operatorname{argmax}} P(M) \end{aligned} \quad (26)$$

where $\hat{M}$ is the position offset of the symmetry center and $\otimes$ is the convolution operator. Thus, translational velocity

Algorithm 1: Procedure of the Proposed Method.

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- Step 1: Calculate the HAF entropy via (16) and obtain the estimation of the micromotion period $\hat{T}_m$.
- Step 2: Calculate the third-order AF of $s(t)$ with the delay time $\tau = \hat{T}_m$ and estimate the jerk via (19). Then, compensate the jerk phase.
- Step 3: Calculate the second-order AF of $s'(t)$ with the delay time $\tau = \hat{T}_m$ and estimate the acceleration via (23). Then, compensate the acceleration phase.
- Step 4: Perform Fourier transform to the signal $\tilde{s}(t)$. Then, search the position offset of the symmetry center from the frequency spectrum via (26) and estimate velocity via (27).
-

can be estimated with

$$\hat{v}_0 = -4\lambda\hat{M} \cdot \text{prf}/N \quad (27)$$

where prf is the pulse repetition frequency (PRF).

From the abovementioned derivation, the proposed method can be summarized in Algorithm 1 as follows:

IV. PERFORMANCE EVALUATION WITH NUMERICAL RESULTS

In this section, this article focuses on the feasibility and performance of the proposed method for the precession cone-shaped target based on electromagnetic computation data. In our experiment, we use the FEKO software to obtain the electromagnetic computation data, and we add the zero-mean Gaussian white noise on the electromagnetic computation data to simulate the target echoes in the noise circumstance. The radar carrier frequency is 10 GHz, and PRF is 1 KHz. In this article, we estimate the micromotion period by finding the peak location of HAF entropy. Therefore, we need to calculate at least one-period HAF entropy to estimate the period. And according to (18), we require at least two periods' echoes to obtain one-period HAF entropy. In this experiment, the dwell time is set to be 1.5 s, which is two and a half periods. The target height H , center height h , and bottom radius r are 3.6, 1.2, and 0.6 m, respectively. The micromotion parameters are set as $\theta = 7^\circ$ and $\omega_c = 2\pi \text{ rad/s}$.

A. Case of Cubic Translation

In the experiment, translational parameters are set as follows: translational velocity $v_0 = 3 \text{ m/s}$, acceleration $a_1 = 1.8 \text{ m/s}^2$, and jerk $a_2 = 2 \text{ m/s}^3$. The SNR is set to be 3 dB. According to [3], the scattering center P_3 is sheltered when the precession angle $0^\circ < \gamma < 90^\circ$. And, when $90^\circ < \gamma < 180^\circ$, the three scattering centers are all observed. First, the simulated precession angle γ is set to be 50° . In this situation, scattering centers P_1 and P_2 are observed. The compensated results of the jerk are given in Fig. 2, and the TF spectrum of the cone-shaped target with cubic translation

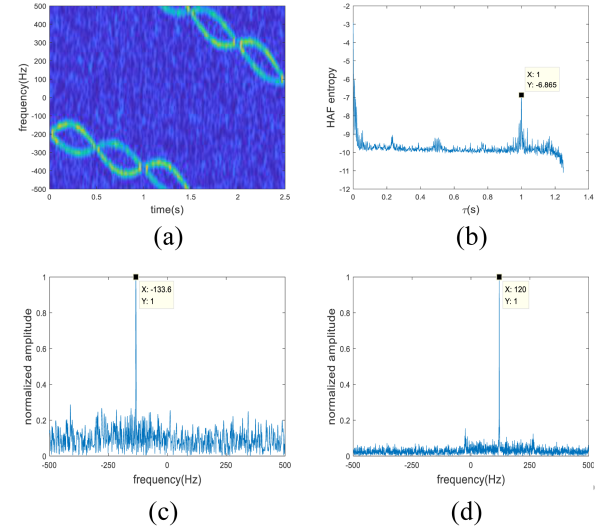


Fig. 2. Estimated jerk and acceleration. (a) TF spectrum of the echo. (b) HAF entropy. (c) Slice of the third-order AF at $\tau = 1 \text{ s}$. (d) Slice of the second-order AF at $\tau = 1 \text{ s}$.

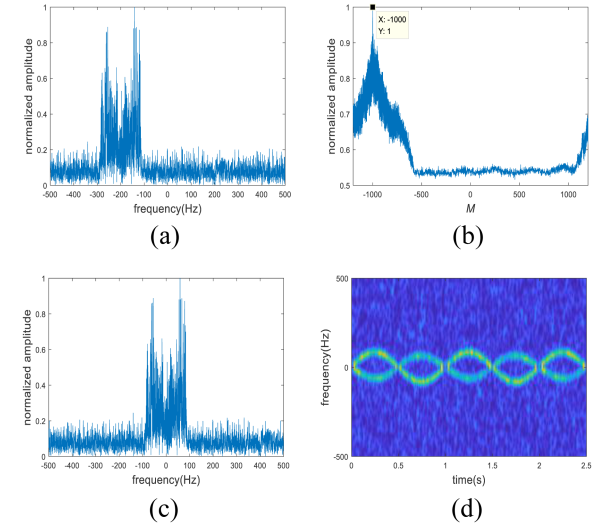


Fig. 3. Velocity compensation results. (a) Frequency spectrum of echo after jerk and acceleration compensation. (b) $P(M)$. (c) Frequency spectrum of echo with complete translation compensation. (d) TF spectrum of echo with complete translation compensation.

is presented in Fig. 2(a). Translational motion causes the shifting, tilting, and aliasing of the TF spectrum. The HAF entropy with the delay time $\tau = \hat{T}_m$ is illustrated in Fig. 2(b). We can see that the peak HAF entropy is $\tau = 1 \text{ s}$, which is almost equal to the real micromotion period $T_m = 1 \text{ s}$. The slice of the third-order AF $S_3(f, \tau)$ at $\tau = 1 \text{ s}$ is displayed in Fig. 2(c), and the jerk can be estimated as $\hat{a}_2 = 2 \text{ m/s}^3$ by using (19). The slice of the second-order AF $S'_2(f, \tau)$ at $\tau = 1 \text{ s}$ is depicted in Fig. 2(d). Thus, acceleration can be estimated as $\hat{a}_1 = 1.7982 \text{ m/s}^2$ by using (23).

Based on jerk and acceleration compensation, Fig. 3(a) presents the frequency spectrum of echo, and Fig. 3(b) plots $P(M)$ in (26). According to (27), velocity can be estimated as $\hat{v}_0 = 3 \text{ m/s}$. Thus far, the micromotion period and all translational parameters are estimated accurately.

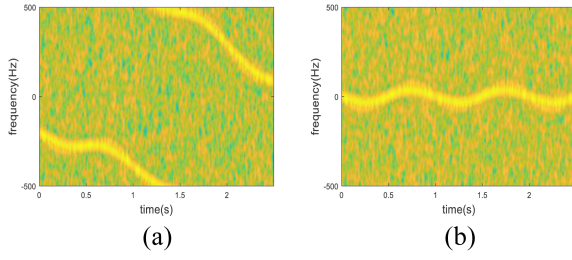


Fig. 4. Experimental results when $\gamma = 130^\circ$. (a) TF spectrum of the echo before translation compensation. (b) TF spectrum of echo with complete translation compensation.

Fig. 3(c) and (d) illustrates the frequency and TF spectrum of echo with complete translation compensation, respectively. As shown in Fig. 3(a) and (c), translational velocity only causes a circular shift of the spectrum along the frequency dimension, which is consistent with the analysis in Section III. This example indicates the effectiveness of our method in cubic translation compensation.

Then, we set $\gamma = 130^\circ$ to simulate a more complex situation, in which three scattering centers are observed. The compensated results of the jerk are given in Fig. 4, and the TF spectrum of the cone-shaped target with cubic translation is presented in Fig. 4(a). Since the scattering coefficient of the scattering center P_1 is obviously lower than those of other scattering centers. The TF spectrum of the scattering center P_1 is drowned in the noise. The parameter estimations are given as follows: $\hat{T}_m = 0.998$ s, $\hat{a}_2 = 1.996$ m/s³, $\hat{a}_1 = 1.8018$ m/s², and $\hat{v}_0 = 2.997$ m/s. Fig. 4(b) shows the TF spectrum of echo with complete translation compensation. Thus far, the micromotion period and all translational parameters are estimated accurately.

B. Case of Quadratic Translation

In this experiment, translational parameters are set as follows: translational velocity $v_0 = 2$ m/s and acceleration $a_1 = 1.5$ m/s². For comparison, we also use Radon [11], LSC [12], and DM methods [13] to realize translational motion compensation. The relative root mean square error (RRMSE) is used in the experiments as a quantitative evaluation criterion, and 300 times Monte Carlo simulations are performed to calculate the RRMSE, which is defined as

$$\text{RRMSE} = \sqrt{\frac{\sum_{n=1}^N (a - \hat{a}_n)^2}{Na^2}} \quad (28)$$

where a is the true translational parameter, $\hat{a}_n$ is the estimated parameter in the n th realization, and N is the number of Monte Carlo simulations. The RRMSE of acceleration and jerk are provided in Tables I and II, respectively. The proposed method has a superior estimation performance to other methods. As a method based on the TF spectrum, the Radon method requires a high SNR and the spectrum resolution limits its accuracy. Thus, the Radon method has the worst estimation accuracy. The DM method is highly dependent on the accurate estimation of the micromotion period. In a low SNR, the results of estimating acceleration

TABLE I
RRMSE of Acceleration Versus SNR ($\times 10^{-3}$ m/s²)

SNR	−4 dB	−2 dB	0 dB	2 dB
Proposed method	13.70	3.151	1.403	1.143
Radon method	683.9	470.3	368.3	308.2
LSC method	32.65	28.46	20.49	20.02
DM method	23.54	22.11	15.94	10.67

TABLE II
RRMSE of Velocity Versus SNR ($\times 10^{-3}$ m/s)

SNR	−4 dB	−2 dB	0 dB	2 dB
Proposed method	1.580	1.506	1.446	1.417
Radon method	468.2	461.3	458.8	457.4
LSC method	15.31	10.13	9.975	9.961
DM method	461.2	459.6	456.6	451.7

are worse because of the estimated period with large errors. The DM method cannot provide accurate estimation results of velocity because of the micro-motion model mismatch.

V. CONCLUSION

In this article, a method of high-order translation compensation for a precession cone-shaped target is proposed. The HAF of the periodical polynomial phase signal is proven periodical along the lag axis and reaches the peak concentration when the lag is equal to the micro-motion period. The micro-motion period is accurately estimated in the case of high-order translation. Since the support area of frequency spectrum of (24) is symmetric around the Doppler shift frequency, we use a convolution operator to find the symmetry center and estimate the velocity. Experimental results based on electromagnetic computation data indicate that the proposed method can realize high-precision translation compensation. In future studies, the proposed method is also applicable for other spinning targets and can be extended to deal with translation with polynomial terms larger than three orders in principle.

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